Original Instructions

FactoryTalk Analytics GuardianAI Quick Installation Guide

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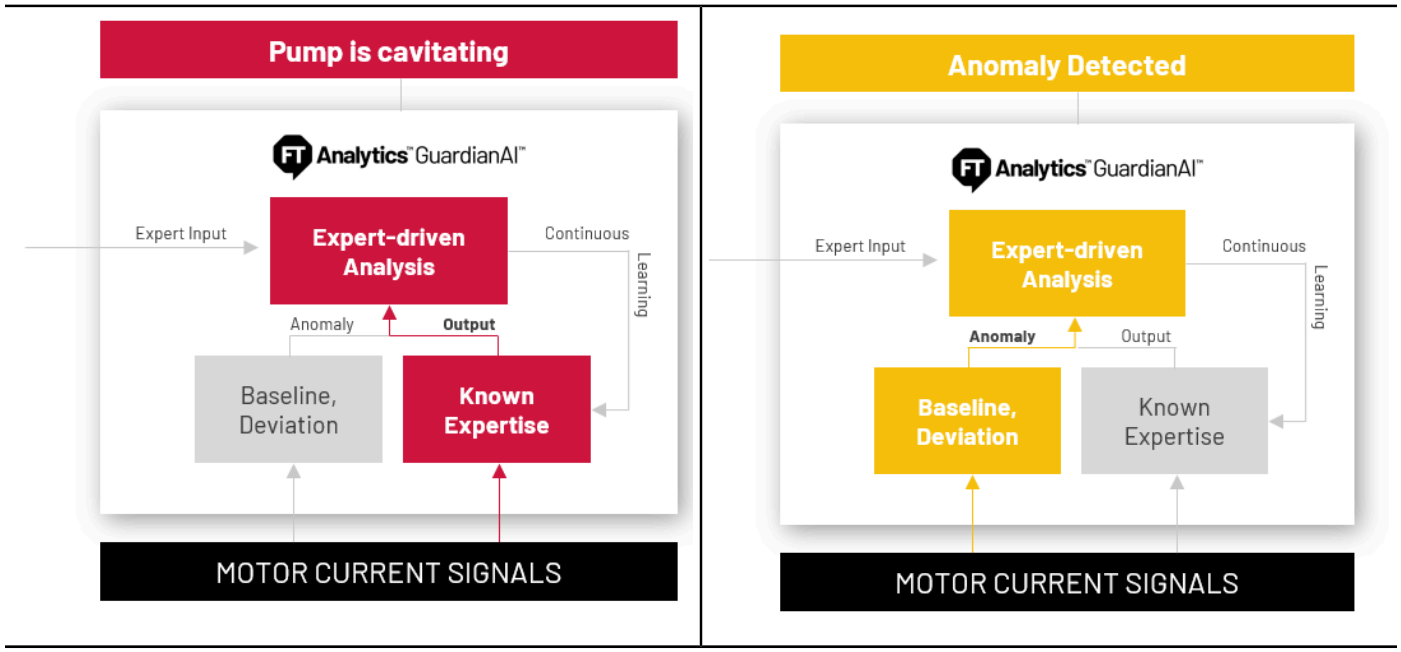
Overview

FactoryTalk® Analytics™ GuardianAI™ is a machine learning based supervisory application that uses existing plant devices, such as variable-frequency drives, as sensors to monitor the health of components such as pumps, fans, and blowers on a plant floor. It uses device data to establish a baseline signature of each component's behavior under normal operating conditions. Then, it monitors the components for any deviation from baseline. Once a deviation is detected, a notification is sent to the user identifying the anomaly. If an anomaly is detected but cannot be identified, FactoryTalk Analytics GuardianAI notifies the maintenance engineer that an unidentified anomaly was detected. The engineer can then investigate the issue, determine the cause of the anomaly, and tag the deviation accordingly. FactoryTalk Analytics GuardianAI machine learning engine then trains to identify the new anomaly for future encounters. The following diagram illustrates this process and the variation between a known fault and an unknown deviation.



The FactoryTalk Analytics GuardianAI workflow takes a no-code approach to machine learning. As a result, a data scientist is not required to configure, deploy, or use this AI application. It is designed so that OT personnel, such as maintenance engineers, controls engineers, machine operators, and plant managers, can work with FactoryTalk Analytics GuardianAI with minimal training required.

The configuration workflow consists of four steps.

1. Deploy the FactoryTalk Analytics GuardianAI application on an ASEM 6300 industrial computer or a local virtual machine.
2. Add the device that will act as a sensor.
3. Provide the identifying information about the component being monitored (pump, fan, blower, or motor).
4. Start training the model to establish the baseline.

FactoryTalk Analytics GuardianAI supports monitoring multiple baselines based on a component's state. Components such as pumps, fans, and blowers on a plant floor can operate under different conditions and processes. FactoryTalk Analytics GuardianAI provides the capability to monitor these varying situations.

NOTE: State refers to a component's specific operating condition, such as different pressures, flow rates, speed, etc.

FactoryTalk Analytics GuardianAI provides premium integration with the controller to seamlessly read changes in the component's tags selected during configuration. When a component's tag value in the controller changes, FactoryTalk Analytics GuardianAI automatically creates a new state and starts acquiring the baseline for that specific state. Once the baseline is acquired, FactoryTalk Analytics GuardianAI continuously monitors the component for any deviation in that particular state. FactoryTalk Analytics GuardianAI will automatically switch to the corresponding state based on the change in the controller tags to ensure accurate monitoring.

For example, if a pump operates under different pressures or flow rates, FactoryTalk Analytics GuardianAI will create and monitor baselines for each condition and switch to the appropriate state when the component's operating condition changes to ensure accurate monitoring.

FactoryTalk Analytics GuardianAI provides premier integration with PowerFlex® 755, 755TL, 755HiHP, 755TR, 755TM, 755TS, and 6000T drives to use as sensors to access three-phase current data for motor current signature analysis. It will focus on anomaly detection and identification for the following component types: pumps, fans, and blowers. The application is designed to work with single-drive and motor applications. Given its adaptive nature, FactoryTalk Analytics GuardianAI can learn process-centric issues and adapt to component types beyond those listed above. For this use case, the application comes equipped with generic motor control analytics.

Container Deployment

This section provides information about deploying FactoryTalk Analytics GuardianAI manually using container.

Pre-requisites

The following are the pre-requisites to install FactoryTalk Analytics GuardianAI:

- Minimum Hardware Specifications:
  - Processor: Intel Atom 4 core
  - RAM: 8 GB
  - SSD: 120 GB
- Recommended Host Hardware Specifications for optimal performance if there are more than 5 components to monitor:
  - Processor: Intel i5 and above
  - RAM: 16 GB
  - SSD: 240 GB
- Host Machine Operating System:
  - Linux Debian 11, 12 (or) Ubuntu 22
- Client Computer
  - Operating System: Microsoft Windows
  - Web Browser: Google Chrome or Microsoft Edge
- Ensure the Linux instance has internet access to enable the automated script to download dependent packages.
- The user must have sudo access to execute system commands. For more information on adding a user to the sudoers list, refer to Add a User to the Sudoers List on page 16.

Retrieve and Deploy the FactoryTalk Analytics GuardianAI Application

Perform the following steps to download the FactoryTalk Analytics GuardianAI application:

1. Download the container package from the Product Compatibility and Download Center (PCDC) at rok.auto/pcdc.

   The download file is named '1.04.00-FTA_GuardianAI-DVD.exe'. Execute it to extract the following files:

   - **guardianai-install.sh** - This installation script is used to deploy FactoryTalk Analytics GuardianAI application.
   - **guardianai-uninstall.sh** - This script is required in case you want to uninstall FactoryTalk Analytics GuardianAI application.
   - **guardianai-reset-password.sh** - This script is used to reset the login password for the FactoryTalk Analytics GuardianAI.
   - **guardianai.tar** - The guardianai image includes application files and all other dependent scripts required to run the FactoryTalk Analytics GuardianAI application.
   - **guardianai-redis.tar** - The Redis database image used as the data storage layer for the FactoryTalk Analytics GuardianAI application.
   - **guardianai-nats.tar** - The nats image used to establish communication between ml-engine and guardianAI containers.

- **guardianai-ml-engine.tar** - A machine learning engine image that contains the algorithms used by FactoryTalk Analytics GuardianAI application.
- **guardianai-backup.tar** - This file contains the dependent scripts required to back up the FactoryTalk Analytics GuardianAI application.
- **k8s-pause-3-5.tar** - A prebuilt pause container image used to initialize pod infrastructure and maintain shared namespaces across containers.
- **backup-raw-data.sh**: This file contains the scripts required to take the backup of Raw data.
- **controller-profile-migration.sh**: This file contains the scripts required to migrate the controller paths from the previous version to controller profiles and provides rollback functionality if the migration is unsuccessful.
- **1.04.00_FactoryTalk Analytics GuardianAI_Quick Installation Guide.pdf**: This file provides step-by-step instructions on how to install or migrate the FactoryTalk Analytics GuardianAI application.

2. Move the files to the target Linux machine.

Use **WinSCP** or equivalent tool to transfer to the Linux machine. Verify that all the files are successfully copied to the target machine.

Install FactoryTalk Analytics GuardianAI Application

Perform the following steps to install the FactoryTalk Analytics GuardianAI application:

If the FactoryTalk Analytics GuardianAI V1.03.00 is already installed, skip this section, and refer to the Upgrade on page 13 section for instructions on upgrading to FactoryTalk Analytics GuardianAI V1.04.00.

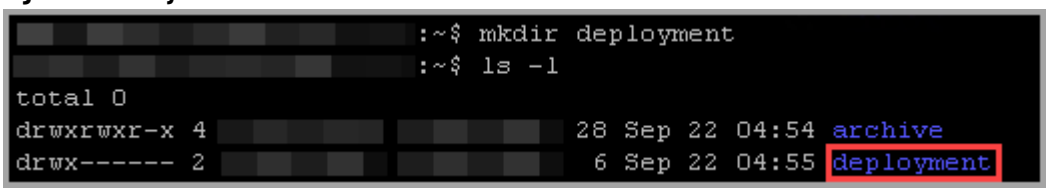
1. Execute the following command to create a directory in **/home/<user>**.

```
mkdir <directory-name>
```

For example, the below command creates a directory named `deployment`.

```
mkdir deployment
```

Figure 1. Directory



NOTE: The `deployment` directory is used as the working directory throughout this document.

2. Move the following files to the `deployment` directory in the Linux machine:
   - guardianai-install.sh
   - guardianai-uninstall.sh
   - guardianai-redis.tar
   - guardianai.tar
   - guardianai-nats.tar
   - guardianai-ml-engine.tar
   - guardianai-backup.tar
   - k8s-pause-3-5.tar
   - guardianai-reset-password.sh
   - backup-raw-data.sh
   - controller-profile-migration.sh
3. Execute the following commands from **/home/<user>** directory to modify the permission to the installation scripts and enable execution:

```
chown -R <user>:<user> deployment
```

```
chmod 755 ./deployment/*.sh
```

4. Navigate to the deployment directory to begin the installation.

```
cd deployment
```

5. Execute the following command to install the FactoryTalk Analytics GuardianAI application:

- The installation script takes the login password as an argument.
- Add the FactoryTalk Analytics GuardianAI application login password to the command before execution.

```
./guardianai-install.sh -p <FactoryTalk Analytics GuardianAI Login Password>
```

NOTE: The user will be prompted to enter the sudo password.

Once the script is executed successfully, the user will see the following:

Figure 2. Installation Script Execution

```
demouser@demovm:~/1.4$ ./guardianai-install.sh -p Rockwell
Starting the setup...
Updating package lists
Installing archive extractor package
Extracting FactoryTalk Analytics GuardianAI package
Extracting database package
Extracting message broker package
Extracting backup package
Extracting ml-engine package
Installing runtime environment
Setting port 443 access
Setting connectivity check from containers
Enabling execution after this session disconnect
Enabling scheduling service, requires super user permission
FactoryTalk Analytics GuardianAI runtime is not installed
FactoryTalk Analytics GuardianAI backup does not exist
FactoryTalk Analytics GuardianAI ml engine does not exist
FactoryTalk Analytics GuardianAI database does not exist
FactoryTalk Analytics GuardianAI does not exist
Loading FactoryTalk Analytics GuardianAI version 1.4.0
Renaming loaded image localhost/1.4.0:latest to localhost/guardianai:1.4.0
FactoryTalk Analytics GuardianAI image localhost/guardianai:1.4.0
Loading FactoryTalk Analytics GuardianAI database version 8.0.0, previous data is retained
Renaming loaded image  localhost/8.0.0:latest to localhost/guardianai/redis:8.0.0
Database image localhost/guardianai/redis:8.0.0
Loading FactoryTalk Analytics GuardianAI message broker version 2.10.24, previous data is retained
Renaming loaded image  localhost/2.10.24:latest to localhost/guardianai/nats:2.10.24
Message Broker  image localhost/guardianai/nats:2.10.24
Loading FactoryTalk Analytics GuardianAI backup version 1.4.0, previous data is retained
Renaming loaded image  localhost/1.4.0:latest to localhost/guardianai/backup_restore:1.4.0
backup & restore image localhost/guardianai/backup_restore:1.4.0
Loading FactoryTalk Analytics GuardianAI ml-engine version 1.4.0, previous data is retained
Renaming loaded image  localhost/1.4.0:latest to localhost/guardianai/ml_engine:1.4.0
ml engine image localhost/guardianai/ml_engine:1.4.0
Creating FactoryTalk Analytics GuardianAI runtime
Creating database
Creating message broker
Creating FactoryTalk Analytics GuardianAI with admin password
Creating FactoryTalk Analytics GuardianAI Backup container
Creating FactoryTalk Analytics GuardianAI ML Engine container
Running FactoryTalk Analytics GuardianAI
Cleaning up FactoryTalk Analytics GuardianAI installation archive
Cleaning up database installation archive
Cleaning up message broker installation archive
Cleaning up backup installation archive
Cleaning up ml-engine installation archive
Start-FactoryTalk Analytics GuardianAI Database Attributes Label / Prescription - 'Shaft Alignment Fault (Duplicate)'
End-FactoryTalk Analytics GuardianAI Database Attributes Label / Prescription - 'Shaft Alignment Fault (Duplicate)'
```

Access FactoryTalk Analytics GuardianAI Application

FactoryTalk Analytics GuardianAI has a self-signed certificate. The user can access the application using Google Chrome or Microsoft Edge.

In a web browser, use the following URL to access FactoryTalk Analytics GuardianAI:

- https://<hostname>

The hostname refers to the instance where FactoryTalk Analytics GuardianAI is installed. Execute the following command to get the instance hostname:

```
hostname -f
```

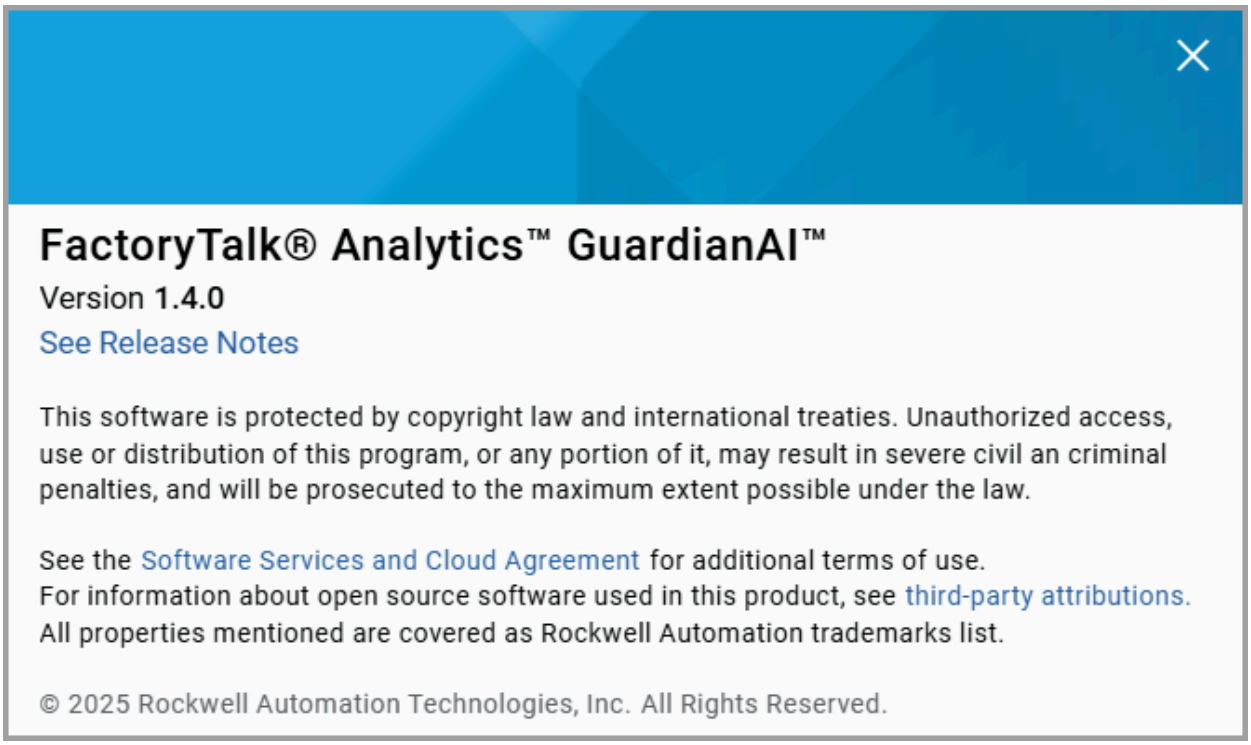
The first access point for FactoryTalk Analytics GuardianAI will be the login page. The login password is the argument set during installation; please refer to step 5 in Install FactoryTalk Analytics GuardianAI Application on page 7.

Figure 3. FactoryTalk Analytics GuardianAI Login Page

After login, the user will be prompted to select the time zone for FactoryTalk Analytics GuardianAI. It is recommended that the time zone be set to the locale of the compute surface where FactoryTalk Analytics GuardianAI is installed. The application uses the time zone to record timestamps of deviation and failure risk events encountered when monitoring the components. The user can modify the time zone on the settings page.

- Click the ⓘ icon in the top-right corner to open the **Help** pane. Click **About Product** to view the version, release notes, license agreement and third-party attributions.

Figure 4. About Product



Backup

The FactoryTalk Analytics GuardianAI application automatically backs up/archives the application data to a volume attached to the **guardianai_backup_restore** container once a day by default. This ensures regular and consistent data backup without any manual intervention.

The backup file is saved with the following name format: **dump-YYYY-MM-DD-hh-mm-ss.guardian**.

Each backup file is considered as one restore point. Each restore point contains the full application data.

Users can change the frequency schedule of backup as required. Perform the following steps to change the backup schedule:

1. Navigate to the `deployment` directory.
2. Execute the following command to change the backup schedule:

   ```
   ./guardianai-install.sh -p <FactoryTalk Analytics GuardianAI Login Password> -b "*/5 * * *
    *"
   ```

   The installation script takes the CRON TIME argument, which will be used to change the application's backup schedule. The above command will re-install the FactoryTalk Analytics GuardianAI application, and the data remains intact in the process.

   NOTE: The user will be prompted to enter the sudo password.

   In the above command, 5 minutes is set as a backup schedule. Every 5 minutes, the FactoryTalk Analytics GuardianAI application automatically backs up the application data.

Restore

Users can restore the application to a specific restore point in the event of any abnormal situations. Perform the following steps to restore the application to a specific restore point:

1. Execute the following command to navigate to the backup and restore container:

```
podman exec -it guardianai_backup_restore sh
```

2. Execute the following command to check the restore points in the index file:

```
cat /data/index
```

The index file contains the list of available restore points. Choose the required restore point and copy the name of the required restore point.

NOTE: The index file is available only when at least one restore point is available.

Figure 5. Index File

```
[redacted]:~$ podman exec -it guardianai_backup_restore sh
/usr/src $ cat /data/index
dump-2024-10-10-11-40-00.guardian
dump-2024-10-10-11-45-00.guardian
dump-2024-10-10-11-50-00.guardian
dump-2024-10-10-11-55-00.guardian
dump-2024-10-10-12-00-00.guardian
```

3. Execute the following command to restore the application to a specific restore point:

```
riot -h $REDIS_HOST -p $REDIS_PORT dump-import -t JSON -z /data/<restore point>
```

- **restore point**: Name of the restore point.

Backup Files Count

A maximum of 30 backup files (restore points) can be stored in **guardianai_backup_restore** container by default. The maximum count can be changed as required.

Perform the following steps to change the maximum backup file count:

1. Navigate to the `deployment` directory.
2. Execute the following command to change the maximum backup file count:

```
./guardianai-install.sh -p <FactoryTalk Analytics GuardianAI Login Password> -c
 <backupcount>
```

The installation script takes the backupcount argument, which will be used to change the maximum backup count. The above command will re-install the FactoryTalk Analytics GuardianAI application, and the data will remain intact.

NOTE: The user will be prompted to enter the sudo password.

Uninstall FactoryTalk Analytics GuardianAI

Perform the following steps to uninstall FactoryTalk Analytics GuardianAI:

NOTE: The uninstall script will delete all the data (training, deviation and failure risks) captured by FactoryTalk Analytics GuardianAI and removes the containerized application.

1. Navigate to the `deployment` directory.
2. Execute the following command to uninstall the FactoryTalk Analytics GuardianAI application.

   ```
   ./guardianai-uninstall.sh
   ```

   Once the script is executed successfully, the user will see the following.

   Figure 6. Uninstall Script Execution

   ```
   [redacted] ./guardianai-uninstall.sh
   Starting the uninstall...
   Disabling FactoryTalk Analytics GuardianAI scheduling service
   Stopping FactoryTalk Analytics GuardianAI runtime
   Removing FactoryTalk Analytics GuardianAI runtime
   Removing installed FactoryTalk Analytics GuardianAI backup localhost/guardianai/backup_restore:1.4.0
   Removing installed FactoryTalk Analytics GuardianAI ml-engine localhost/guardianai/ml_engine:1.4.0
   Removing installed FactoryTalk Analytics GuardianAI database localhost/guardianai/redis:8.0.0
   Removing installed FactoryTalk Analytics GuardianAI message broker localhost/guardianai/nats:2.10.24
   Removing installed FactoryTalk Analytics GuardianAI localhost/guardianai:1.4.0
   Removing FactoryTalk Analytics GuardianAI database files
   Removing FactoryTalk Analytics GuardianAI logs
   Removing FactoryTalk Analytics GuardianAI captured trends
   Removing FactoryTalk Analytics GuardianAI backup captured data
   ```

Reset Application Login Password

In the event a user forgets the login password for the FactoryTalk Analytics GuardianAI application, the following script can be used to reset the password. The login password will be reset to the original password configured during application installation (refer to step 3 below to query the original password).

1. Navigate to the `deployment` directory.
2. Execute the following command to reset the password.

   ```
   ./guardianai-reset-password.sh
   ```

   The script will reset the password to the original value set during the FactoryTalk Analytics GuardianAI installation.

   Figure 7. Password Reset

   ```
   [redacted]:~/deployment$ ./guardianai-reset-password.sh
   Deleting existing password
   The password has been reset to the value that was used during product install
   ```

3. To query the original password that is set during the installation, execute the following command:

   ```
   podman inspect guardianai|grep Admin__Password=
   ```

   Figure 8. Query Password Set During Installation

   ```
   [redacted]:~/deployment$ podman inspect guardianai|grep Admin__Password=
               "Admin__Password=RedBe@kMagner%0!",
               "Admin__Password=RedBe@kMagner%0!",
   ```

   Users can now login using the original password set during installation.

4. To change the password, navigate to the Settings in FactoryTalk Analytics GuardianAI.

Upgrade FactoryTalk Analytics GuardianAI Application

This section provides step-by-step instructions for upgrading the FactoryTalk Analytics GuardianAI application to V1.4.0. The installation script streamlines the installation and seamlessly upgrades the FactoryTalk Analytics GuardianAI application from the previous version while keeping the data intact.

NOTE: Users must be on version 1.3.0 to upgrade to version 1.4.0. Upgrades from earlier versions are not supported.

Perform the following steps to upgrade the FactoryTalk Analytics GuardianAI application from V1.3.0 to V1.4.0:

1. Execute the following command to create a directory with name as **upgrade** in **/home/<user>**.

```
mkdir upgrade
```

2. Move the following files to the **upgrade** directory in the Linux machine:
   - guardianai-install.sh
   - guardianai-uninstall.sh
   - guardianai-redis.tar
   - guardianai.tar
   - guardianai-nats.tar
   - guardianai-ml-engine.tar
   - guardianai-backup.tar
   - k8s-pause-3-5.tar
   - guardianai-reset-password.sh
   - backup-raw-data.sh
   - controller-profile-migration.sh

   IMPORTANT: Verify that all the files listed above are available before continuing the migration.

3. Execute the following commands from **/home/<user>** directory to modify the permission to the installation scripts and enable execution:

```
chown -R <user>:<user> upgrade
```

```
chmod 755 ./upgrade/*.sh
```

4. Navigate to the `upgrade` directory.
5. Execute the following command to upgrade the FactoryTalk Analytics GuardianAI application:

   - The installation script takes the login password and the migration version number as arguments.
   - Add the FactoryTalk Analytics GuardianAI application login password to the command before execution.
   - The -m argument invokes the "backup-raw-data.sh" and "controller-profile-migration.sh" scripts to complete the upgrade process.

```
./guardianai-install.sh -p <FactoryTalk Analytics GuardianAI Login Password> -m 1.4
```

   NOTE: The user will be prompted to enter the sudo password.

The script installs all the necessary dependencies to upgrade the FactoryTalk Analytics GuardianAI application to V1.4.0.

Once the script is executed successfully, the user will see the following example image:

Figure 9. Upgrade FactoryTalk Analytics GuardianAI

```
guardianai@[redacted]:~/[redacted]$ ./guardianai-install.sh -p Rockwell -c 10 -b "*/5 * * * *" -m 1.4
Starting the setup...
Updating package lists
Installing archive extractor package
Extracting FactoryTalk Analytics GuardianAI package
Extracting database package
Extracting message broker package
Extracting backup package
Extracting ml-engine package
Installing runtime environment
Setting port 443 access
Setting connectivity check from containers
Enabling execution after this session disconnect
Enabling scheduling service, requires super user permission
Removing existing FactoryTalk Analytics GuardianAI runtime
Removing installed FactoryTalk Analytics GuardianAI backup localhost/guardianai/backup_restore:1.3.0
FactoryTalk Analytics GuardianAI ml engine does not exist
Removing installed FactoryTalk Analytics GuardianAI database localhost/guardianai/redis:7.2.4
Removing installed FactoryTalk Analytics GuardianAI localhost/guardianai:1.3.0
Loading FactoryTalk Analytics GuardianAI version 1.4.0
Renaming loaded image localhost/1.4.0:latest to localhost/guardianai:1.4.0
FactoryTalk Analytics GuardianAI image localhost/guardianai:1.4.0
Loading FactoryTalk Analytics GuardianAI database version 8.0.0, previous data is retained
Renaming loaded image  localhost/8.0.0:latest to localhost/guardianai/redis:8.0.0
Database image localhost/guardianai/redis:8.0.0
Loading FactoryTalk Analytics GuardianAI message broker version 2.10.24, previous data is retained
Renaming loaded image  localhost/2.10.24:latest to localhost/guardianai/nats:2.10.24
Message Broker  image localhost/guardianai/nats:2.10.24
Loading FactoryTalk Analytics GuardianAI backup version 1.4.0, previous data is retained
Renaming loaded image  localhost/1.4.0:latest to localhost/guardianai/backup_restore:1.4.0
backup & restore image localhost/guardianai/backup_restore:1.4.0
Loading FactoryTalk Analytics GuardianAI ml-engine version 1.4.0, previous data is retained
Renaming loaded image  localhost/1.4.0:latest to localhost/guardianai/ml_engine:1.4.0
ml engine image localhost/guardianai/ml_engine:1.4.0
Creating FactoryTalk Analytics GuardianAI runtime
Creating database
Creating message broker
Creating FactoryTalk Analytics GuardianAI with admin password
Creating FactoryTalk Analytics GuardianAI Backup container
Creating FactoryTalk Analytics GuardianAI ML Engine container
guardianai
Migrating Redis data from 1.3 to 1.4
Step 1: Backing up Redis hashes to '/app/backups/redis'…
↳ Backup complete

Step 2: Delete Outdated data entries…
Deleting keys matching 'GuardianModelService.Model*' from Redis…
1
Deleted: GuardianModelService.Model.Deviations.0.e300550b-eb09-4b2d-adcb-ffaf5511eee9
1
Deleted: GuardianModelService.Model.State.e300550b-eb09-4b2d-adcb-ffaf5511eee9
↳ All matching keys removed.

Step 3: Migrate ControllerProfile entries…
4
  ↳ Created GuardianAssetService.ControllerProfile.fae5605b-e7f7-4d89-a263-1563e727d297 → Name='Demo Folder' Path=[redacted]
ControllerProfile migration has been completed successfully.
Patching SecurityService.AdminUser.AdminUser with ApplicationURL field…
1
↳ ApplicationURL field added to SecurityService.AdminUser.AdminUser

Step 5: Augment AssetInformation hashes…
  ☐ Patching GuardianAssetService.AssetInformation.e300550b-eb09-4b2d-adcb-ffaf5511eee9
0
1
1
1
1
1
1
1
↳ AssetInformation augmentation complete

Upgrade to GuardianAI 1.4: ALL STEPS FINISHED ✓
Running FactoryTalk Analytics GuardianAI
Cleaning up FactoryTalk Analytics GuardianAI installation archive
Cleaning up database installation archive
Cleaning up message broker installation archive
Cleaning up backup installation archive
Cleaning up ml-engine installation archive
Start-FactoryTalk Analytics GuardianAI Database Attributes Label / Prescription - 'Shaft Alignment Fault (Duplicate)'
End-FactoryTalk Analytics GuardianAI Database Attributes Label / Prescription - 'Shaft Alignment Fault (Duplicate)'
```

NOTE: The install script will take care of the following steps during the upgrade:

a. **Redis Upgrade**: Upgrades the Redis database to version 8.0.0.
b. **Data Backup**: Backs up Redis hashes to `/app/backups/redis` before migration.
c. **Data Cleanup**: Removes obsolete data based on predefined rules .
d. **Controller Profile Creation**: Automatically creates controller profiles if controller paths exist in the previous release. Profile names are derived from the folder names containing the controller connection .
e. **Component Hash Augmentation**: Updates existing component hashes to reflect changes in the corresponding components .

IMPORTANT: Following the upgrade, training must be performed again for each component to attain baselines and resume monitoring. Note that only raw data is available for each component.

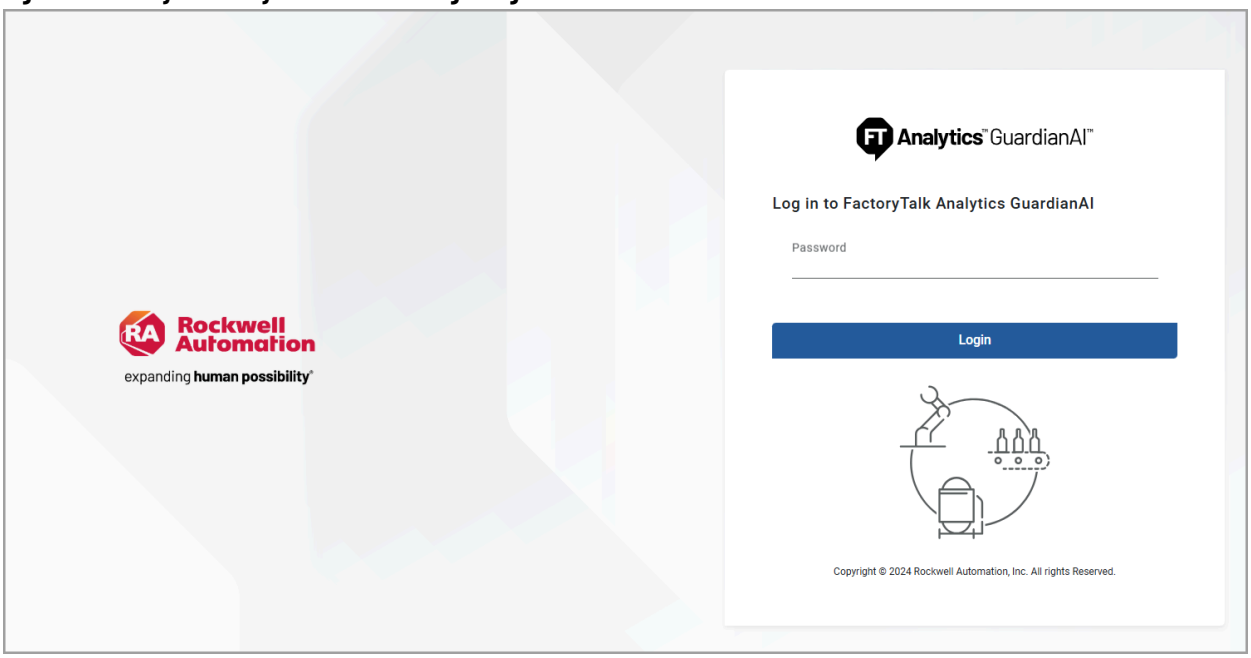
6. After a successful upgrade, clear the browser cache before accessing the FactoryTalk Analytics GuardianAI application user interface.

NOTE: If migration is unsuccessful, the installation script will automatically rollback the application to V 1.3.0. Existing data remains intact. Contact Rockwell Automation support for any difficulties during the upgrade.

7. To verify, use the following URL to access FactoryTalk Analytics GuardianAI:

   https://<hostname>

Figure 10. FactoryTalk Analytics GuardianAI Login Page



The login page is displayed as shown above.

Add a User to the Sudoers List

Adding a user to the sudoers list allows to perform administrative tasks without logging in as root. Always adhere to security best practices before granting application installation rights or adding users to the sudoers list, as these actions can significantly impact system integrity and access control. Refer to the Security Considerations on page 18 given in Appendix.

To add a user to the sudoers list, do the following steps :

NOTE: A root user must execute this process.

1. To switch to the root user, complete the following steps:
   a. Execute the following command:
   ```
   sudo su
   ```
   b. Provide the root password.
2. To add a user to the sudoers list, complete the following steps:
   a. Execute the following command:
   ```
   sudo visudo
   ```
   The visudo command ensures that you do not make syntax errors that can lock you out of the system.

   Figure 11. sudoers.tmp
   ```
   GNU nano 5.4                    /etc/sudoers.tmp
   #
   # This file MUST be edited with the 'visudo' command as root.
   #
   # Please consider adding local content in /etc/sudoers.d/ instead of
   # directly modifying this file.
   #
   # See the man page for details on how to write a sudoers file.
   #
   Defaults        env_reset
   Defaults        mail_badpass
   Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:

   # Host alias specification

   # User alias specification

   # Cmnd alias specification

   # User privilege specification
   root    ALL=(ALL:ALL) ALL
   ```
   b. Use the Down arrow key and scroll down to the line below `root ALL=(ALL:ALL) ALL`.
   c. Add `<Username> ALL=(ALL) ALL`.

   Figure 12. Add User
   ```
   # User privilege specification
   root    ALL=(ALL:ALL) ALL
   guardianaiuser ALL=(ALL) ALL
   ```
3. To exit, press **CTRL + Y**.
4. To save, type the letter **Y**.

Figure 13. Save Sudoers List

```
  GNU nano 5.4                    /etc/sudoers.tmp *
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:>

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
guardianaiuser ALL=(ALL) ALL
# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

File Name to Write: /etc/sudoers.tmp
^G Help              M-D DOS Format       M-A Append           M-B Backup File
^C Cancel            M-M Mac Format       M-P Prepend          ^T Browse
```

5. To write to **sudoers.tmp** file, press **Enter** again.
6. Execute the following command to exit from the root and switch back to the user:

```
exit
```

Figure 14. Exit

```
root@guardianaivm:/home/guardianaiuser# exit
exit
guardianaiuser@guardianaivm:~$
```

Security Considerations

This provides information about security considerations for FactoryTalk Analytics GuardianAI.

When deploying FactoryTalk Analytics GuardianAI within operations, considerations should be applied to the physical and cyber system security posture of the system. The FactoryTalk Analytics GuardianAI application is designed to run on a client supplied edge hardware and infrastructure.

System security is a paramount tenet of overall operation success. The following document and links are being provided as guidance on best practices of how to implement system security principles within industrial automation control systems. It is considered best practice for users of FactoryTalk Analytics GuardianAI to follow these practices to provide the best level of defense in depth security for their overall systems.

Table 1. Security Publications

Publication Name	Description	Link
Converged Plantwide Ethernet (CPwE) Design	Converged Plantwide Ethernet Design Implementation Guide	Converged Plantwide Ethernet Design
Deploying Firewalls within CPwE Architecture	Use cases for designing, deploying, and managing industrial firewalls	Deploying Firewalls within CPwE
Deploy Identity and Mobility Services	Guidelines for protecting systems through deploying centrally managed defense in-depth security approach	Identity and Mobility Services
Secure Cloud Connectivity to CPwE	Application guide for securing cloud applications within CPwE Architecture	Secure Cloud Connectivity to CPwE
Deploy CIP Security within CPwE	Network security use cases for CPwE systems	Deploy Network Security in CPwE
Deploy CIP Security within CPwE	IEC 62443 security architecture use cases and design principles	Deploy CIP Security in CPwE
Physical Infrastructure within CPwE Architecture	Use cases for deploying robust physical infrastructure for industrial applications	Physical Infrastructure within CPwE Architecture

Network Security and Segmentation

Network security practices within Converged Plantwide Ethernet define the usage of zones and conduits to segment components within an industrial automation control system. When using FactoryTalk Analytics GuardianAI it is suggested to follow the CPwE Design Implementation guide (Converged Plantwide Ethernet Design) to isolate PowerFlex Drives into various zones. Furthermore, augmenting these zones and conduits with enhanced physical security is also a best practice as defined by Physical Infrastructure within CPwE architecture (Physical Infrastructure within CPwE Architecture).

Email Relay Server

One of the means of communication of notifications for FactoryTalk Analytics GuardianAI is a SMTP server for email notification. This SMTP server is to be provided by the user of FactoryTalk Analytics GuardianAI. It is best practice to secure this SMTP server with the defense-in-depth guidance provided by CPwE Design (Converged Plantwide Ethernet Design). Access to the FactoryTalk Analytics GuardianAI application is access-controlled, but the user of FactoryTalk Analytics GuardianAI is responsible for the security of the SMTP server.

Access to the FactoryTalk Analytics GuardianAI Application

Access to the FactoryTalk Analytics GuardianAI is privileged and access-controlled. It is up to the user of FactoryTalk Analytics GuardianAI to use strong passwords and to protect these passwords. It is also the responsibility of the user to provide appropriate access controls to the IP addresses of the applications within the facility where the application is running.

Data Integrity and Confidentiality

There is no explicit data or configuration that is shared with FactoryTalk Analytics GuardianAI. The only information exchanged between the PowerFlex Drive and FactoryTalk Analytics GuardianAI application is the three electrical phase currents and the electrical frequency. The user of the FactoryTalk Analytics GuardianAI application is responsible for protecting the container runtime environment.

Denial of Service

The FactoryTalk Analytics GuardianAI application is a small edge run application that only monitors information from PowerFlex drives to provide users insights about electromechanical anomalies and deviations. The application is intended to be run only on internal networks protected by defense-in-depth security as defined by CPwE (Converged Plantwide Ethernet Design). There is no intelligent load balancer built into this application. The user is responsible for guarding against denial-of-service attacks.